

AKSHAYA JAYAKANTH

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EDUCATION

Brown University, Providence, RI, ScM in Data Science

Aug 2025 – May 2027 (expected)

Coursework: Data Engineering, Statistical Learning, Hands-on Data Science, Fairness in Automated Decision Making. Cross-registered at Harvard University (Fall 2026): AC215 MLOps and LLMOps: Production AI Systems; Advanced Practical Data Science.

SRM Institute of Science and Technology, Chennai, India, B.Tech in CSE, Big Data Analytics

Sept 2021 – June 2025

PROFESSIONAL EXPERIENCE

Annenberg Institute at Brown University, Text-as-Data for Program Evaluation (TEAM)

Jun 2026 – Present

Graduate Research Assistant, internal hire after capstone; Connecticut, Indiana, and California records

Sept 2026 – Present

NLP Capstone Researcher, Connecticut TEAM records

Jun 2026 – Aug 2026

- Run the preprocessing and embedding pipeline for 100,000+ Connecticut TEAM administrative records (growth plans, reflection papers, performance notes) as **SLURM** batch jobs on **OSCAR HPC**, Brown's Linux cluster; validate the schema of every new raw export against sample data before it enters the pipeline, so schema drift is caught before ingestion rather than downstream.
- Run and interpret **BERTopic** topic models on **sentence-transformer embeddings**, chosen after benchmarking them against **TF-IDF/SVD**; QC cleaning and embedding-validation reports, flag anomalies, review team scripts for efficiency, and maintain reproducible documentation of logs, SLURM submissions, and pipeline decisions in **GitHub**.

Lizzie and Jonathan Tisch Center for Career Exploration, Brown University, Student Research Assistant

Jun 2026 – Aug 2026

- Designed the **data architecture** and unified **relational schema** for the Career Readiness at Brown (CR@B) initiative, integrating 5 years of engagement and outcome data (undergraduates to alumni, every department) into an access-controlled **PostgreSQL** repository; diagnosed a wide-vs-long mismatch in the Handshake export and re-architected the **dbt** staging melt, eliminating cross-dataset join errors.
- Built automated **ETL pipelines** in **Python** and **SQL** on **OSCAR HPC** with **PII pseudonymization** (**spaCy NER**, **UUID5** crosswalk) and **synthetic data generation** (**Faker**, **SDV**); shipped through PR review with Ruff linting, **unittest** suites, and **CI** coverage gates, and connected PostgreSQL to **Tableau** for stakeholder dashboards and downstream **predictive modeling**.

In22labs, AI/ML Intern, Chennai, India (Hybrid)

Aug 2024 – Jan 2025

- Replaced OpenAI API calls in a document QA service with self-hosted open-source models at 2.7 s median response latency: built a **FastAPI** ingestion API with **OCR** for PDF, TXT, PNG, and JPG, and benchmarked **Llama 3.2**, **Gemma**, and **Mistral** via **Ollama** on retrieval accuracy and latency.
- Negotiated an In22labs and SRM University MoU and onboarded 17 interns across 2 batches during final year of undergraduate study.

Agastya Data Solutions, Data Science Intern, Hyderabad, India

Jun 2023

- Built a holding-period return prediction tool on 5 years of NIFTY 50 constituent data (**Python**, **SQL**): **data cleaning**, **exploratory analysis**, and visual reports for investment decisions; the dataset became the basis for the DRL portfolio paper below.

PROJECTS

- Equity Portfolio Analysis Tool (Python, Streamlit)**: Built a live **yfinance** ingestion **data pipeline** over roughly 1,500 trading days for 50+ stocks; implemented momentum-based selection and benchmarked risk-adjusted returns via CAGR, volatility, and **Sharpe ratio**.
- In-Hospital Mortality Prediction, SUPPORT2 Dataset (Python, Scikit-learn)**: Reached 0.916 AUC with **Gradient Boosting** on **clinical records** for 9,105 critically ill patients across 42 features after **data preprocessing** and comparing five models under 5-fold stratified **cross-validation**; **SHAP** identified acute physiology severity (avtisst) as the top predictor.
- Early Prediction of Movie Box Office Success from Wikipedia Activity (R)**: Predicted revenue one month pre-release at cross-validated R^2 of 0.78 for 312 films by **feature engineering** on time-aware Wikipedia activity signals and fitting **linear regression** and **Gamma GLMs** under 10-fold **cross-validation**, with log-log transforms on skewed revenue; page views and theater count were the dominant predictors.

PUBLICATIONS AND PATENT

- AI-Driven Liver Disease Diagnosis: Multi-Modal Fusion of Medical Imaging and Clinical Reports (Patent No. 202541059774)**: Multi-modal system fusing EHR data, synthetic medical imaging, and NLP pipelines for liver disease detection; **Vision Transformers** in place of CNNs; 85–93% precision.
- Enhancing Portfolio Optimisation with Deep Reinforcement Learning (IJSREM)**: **DQN** and policy-gradient (**PPO**, **TRPO**) agents on NIFTY50 data with Sharpe ratio as reward; beat Mean-Variance Optimisation on risk-adjusted returns in volatile regimes. [\[link\]](#)
- Real-Time Personality Assessment via Big Five Traits (Research Square)**: NLP pipeline extracting Big Five traits from text via **embeddings**, sentiment scoring, and behavioral features; trained on a 4GB corpus for HR screening. [\[link\]](#)

SKILLS

Programming: Python, SQL (window functions, joins, normalization), R, Bash, Git, Linux

Data Engineering: ETL design, PostgreSQL, dbt, schema design, data modeling, PII pseudonymization, synthetic data (Faker, SDV)

Cloud and MLOps: Docker, Docker Compose, Google Cloud Platform (Cloud Storage, Cloud Run, Vertex AI), DVC, Weights and Biases, LangChain, RAG, LoRA fine-tuning, quantization, Kubernetes, CI/CD

Testing, HPC and Visualization: unittest, Ruff, CI/CD (coverage gates), OSCAR HPC, Slurm, Tableau, Matplotlib

ML and NLP: Scikit-learn, TensorFlow, PyTorch, spaCy, Sentence-Transformers, BERTopic, SHAP, GLMs, cross-validation, deep reinforcement learning

LLMs and Serving: Ollama (Llama 3.2, Gemma, Mistral), HuggingFace, FastAPI, REST APIs, Streamlit, OCR pipelines

HONORS AND LEADERSHIP

Founder, Teen Clean (10+ years), nonprofit expanding sanitation access for girls in rural communities. Awards: Phenomenal She; Abdul Kalam; Young Social Worker; Overall Proficiency, 1st Rank.